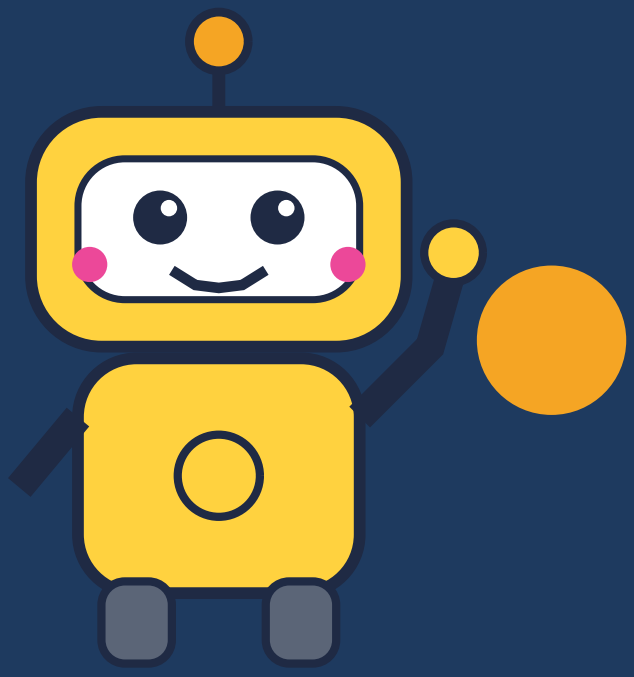
SAMPLE PAPERS

Think Like a Coder

Sample examination papers, Class 5

Half-yearly and annual papers
40 marks, 1½ hours, with blueprints and marking schemes



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 5 · Photocopiable for school use

SAMPLE EDITION · DRAFT

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About these papers

These are two sample examination papers for Class 5: a **half-yearly** paper on the term-1 chapters and an **annual** paper on the whole year. They go with *Think Like a Coder*, Class 5, and with the chapter assessments. All questions are new.

How they follow CBSE

- CBSE does not hold board examinations for Classes 3-8 and has not published a question-paper pattern for CT&AI.; Schools set their own papers. These papers follow what CBSE's CT&AI; handbooks describe: **competency-based** written tests with CT puzzles, with no 'write a program' questions and no computer needed.
- The question types are the ones CBSE uses in its competency-based papers: multiple choice, short answers and case-based questions.
- Questions are spread across four thinking levels (remember, understand, apply, analyse and create), with most marks for applying and analysing. The blueprints show the split.
- Chapters follow the order of CBSE's 2026-27 CT&AI; handbooks and the NCERT maths chapters.

Paper design

Section	Question type	Questions	Marks
A	Multiple choice	8	8 x 1 = 8
B	Very short answer	6	6 x 2 = 12
C	Short answer	4	4 x 3 = 12
D	Case-based questions	2	2 x 4 = 8
	Total · 1½ hours	20	40

Using the papers

- Use the half-yearly paper in September and the annual paper in February-March, or as mock tests before the school's own exams.
- Print one paper per student; the papers have space for answers.
- Read questions aloud to students who find reading hard: the paper tests thinking, not reading.
- Mark with the marking scheme; accept any correct method, and give part marks for correct steps.
- Use the blueprints to set your own papers in the same pattern.

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SAMPLE PAPER · HALF-YEARLY EXAMINATION

Class 5 · Computational Thinking

Syllabus: chapters 1-8 (term 1)

Time allowed: 1½ hours **Maximum marks:** 40

Name: _____ **Section:** _____ **Roll no.:** _____

General instructions

1. This paper has 20 questions in 4 sections. All questions are compulsory.
   Section A: multiple choice, 8 questions of 1 mark each; Section B: very short answer, 6 questions of 2 marks each;
2. Section C: short answer, 4 questions of 3 marks each; Section D: case-based questions, 2 questions of 4 marks each.
3. Marks for each question are shown on the right.
4. Write your answers in the space given below each question.
5. No calculator or computer is needed. Show your working: marks are given for correct steps.

SECTION A · Multiple choice — 8 x 1 = 8 marks

1. 'The sky is blue OR grass is purple.' This statement is ... [1]

a) true **b)** false **c)** can't tell

2. A number is even AND more than 10. Which number fits? [1]

a) 8 **b)** 12 **c)** 15

3. Set cake to 12 slices. Eat one third of them. Now cake is ... [1]

a) 4 **b)** 8 **c)** 9

4. A right angle is ... [1]

a) 45° **b)** 90° **c)** 180°

5. A turtle draws a triangle with equal sides. Each turn is ... [1]

a) 60° **b)** 90° **c)** 120°

6. A program asks 'How many km?' The number you type in is the program's ... [1]

a) input **b)** output **c)** loop

7. A running total starts at 0. Add 4, then 6, then 5. The total is ... [1]

a) 10 **b)** 15 **c)** 16

8. Which search is usually faster on a long list in order? [1]

a) one-by-one search **b)** halving search

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SECTION B · Very short answer — 6 x 2 = 12 marks

9. Find 3/4 of 24, and 2/5 of 20. [2]

10. The turtle faces North and turns 270° clockwise. Which way does it face now? [2]

11. Write >, < or =: (a) 2 km ___ 1,999 m (b) 4,000 m ___ 4 km [2]

12. In a picture graph one jug means 4 litres. How many litres do 3½ jugs show? How many jugs show 10 litres? [2]

13. The rule is dots = 4 x step - 1. How many dots are there in step 1, and in step 10? [2]

14. A program says: set fuel to 30; fuel = fuel + 12; fuel = fuel - 20. What is fuel at the end? Which line is the refill? [2]

SECTION C · Short answer — 4 x 3 = 12 marks

15. A 2-digit number has digits that add up to 5. Is each statement true, false or can't tell? (a) It is less than 60. (b) It is odd. (c) It has a 9 in it. [3]

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16. A secret number is from 1 to 100. It is 70. Write the first three guesses of a halving search, and which numbers are left after each. [3]

17. Step 1 has 1 tile, step 2 has 4 tiles and step 3 has 9 tiles (squares of tiles). Write the rule. How many tiles in step 8? Which step has 100 tiles? [3]

18. Set nearest to 1,000 m. Check four shops: 450 m, 1 km 200 m, 380 m and 600 m. If a shop is nearer, update nearest. Write nearest after each check. Which shop is nearest? [3]

SECTION D · Case-based questions — 2 x 4 = 8 marks

19. Five cows give 9, 11, 8, 12 and 10 litres of milk. A program keeps a running total and a counter.

(a) Write the running total after each cow. [2]

(b) What does the counter show at the end? [1]

(c) In a picture graph one cup = 10 litres. How many cups show the total? [1]

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20. A school bus program uses a variable called children. It starts at 0. At stop 1, 12 children get on. At stop 2, 8 get on and 3 get off. At school, everyone gets off.

(a) What is children after stop 1? [1]

(b) What is children after stop 2? [1]

(c) What is children after school? [1]

(d) The bus has 40 seats. Write an if-else rule the program could use at each stop. [1]

***** End of the question paper *****

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SAMPLE PAPER · ANNUAL EXAMINATION

Class 5 · Computational Thinking

Syllabus: all chapters, mostly chapters 9-15 (term 2)

Time allowed: 1½ hours **Maximum marks:** 40

Name: _____ **Section:** _____ **Roll no.:** _____

General instructions

1. This paper has 20 questions in 4 sections. All questions are compulsory.
   Section A: multiple choice, 8 questions of 1 mark each; Section B: very short answer, 6 questions of 2 marks each;
2. Section C: short answer, 4 questions of 3 marks each; Section D: case-based questions, 2 questions of 4 marks each.
3. Marks for each question are shown on the right.
4. Write your answers in the space given below each question.
5. No calculator or computer is needed. Show your working: marks are given for correct steps.

SECTION A · Multiple choice — 8 x 1 = 8 marks

1. A machine's rule is + 7. Input 15 gives ... [1]

a) 8 **b)** 22 **c)** 105

2. A design with 8 petals turns by how much after each petal? [1]

a) 40° **b)** 45° **c)** 60°

3. Covering a floor with one repeating tile and no gaps is called ... [1]

a) tiling **b)** sorting **c)** timing

4. How many seconds are there in 2 minutes? [1]

a) 60 **b)** 100 **c)** 120

5. A frog starts at 0 and jumps 6 each time. Which spot does it land on? [1]

a) 16 **b)** 30 **c)** 40

6. A map scale says 1 square = 50 m. How far are 4 squares? [1]

a) 54 m **b)** 200 m **c)** 450 m

7. Which graph uses small pictures to show data? [1]

a) a picture graph **b)** a bar graph **c)** a number line

8. The turtle turns right 90° four times. Now it is ... [1]

a) facing the same way as at the start **b)** facing the opposite way

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SECTION B · Very short answer — 6 x 2 = 12 marks

9. A machine turns 4 into 13 and 6 into 19. Find the rule. [2]

10. What is the order of rotational symmetry of a square? Of a triangle with equal sides? [2]

11. A shape covers 9 whole squares and 6 half squares. What is its area? [2]

12. Race times: Neha 1 minute 5 seconds, Om 58 seconds, Pia 62 seconds. Write all the times in seconds. Who won? [2]

13. A 2-jumper and a 5-jumper start at 0. Where do they first land on the same spot? Where is the next one? [2]

14. Start at C2. Go North 2, then East 3. Where are you now? [2]

SECTION C · Short answer — 4 x 3 = 12 marks

15. Machine A doubles a number (x 2). Machine B takes away 3 (- 3). (a) Put 10 through A, then B. (b) Put 10 through B, then A. (c) Does the order matter? [3]

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16. Set timer to 10. Repeat until timer = 0: say timer, then timer = timer - 3. Trace it. What is the bug? Fix it. [3]

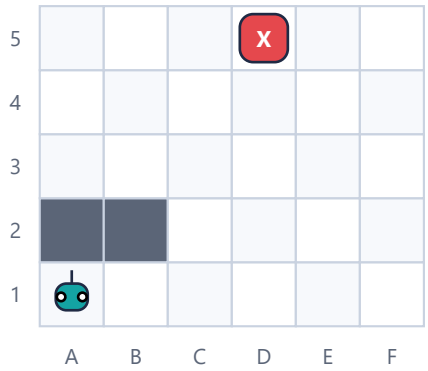
17. Set position to 0. Repeat until position > 40: position = position + 3; if position is a multiple of 5 then say 'Meet!'. At which positions does it say 'Meet!'? [3]

18. A floor 6 squares long and 4 squares wide is covered with 2 x 2 tiles. Each tile has 1 red square and 3 white squares. How many tiles, red squares and white squares? [3]

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SECTION D · Case-based questions — 2 x 4 = 8 marks

19. Robo starts on A1. The grey squares are walls.





(a) Where is the treasure X? [1]

(b) Write a shortest route from A1 to X with compass directions, avoiding the walls. [2]

(c) 1 square = 100 m. How far is your route? [1]

20. A treasure game scores: +10 for the treasure, +2 for each coin, -3 for each wall bump. Ana found the treasure, 3 coins and bumped 2 walls. Bo found the treasure and 1 coin, with no bumps.

(a) What is Ana's score? [1]

(b) What is Bo's score? [1]

(c) Who wins? [1]

(d) Suggest one change to make the scoring fairer, and say why. [1]

***** End of the question paper *****

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Blueprints

Blueprint: Half-yearly examination

Syllabus: chapters 1-8 (term 1). Marks by chapter and section.

Chapter	A	B	C	D	Marks
1. We the Travellers - I	2		3		5
2. Fractions	1	2			3
3. Angles as Turns	2	2			4
4. We the Travellers - II	1	2		4	7
5. Far and Near		2	3		5
6. The Dairy Farm	1	2		4	7
7. Shapes and Patterns		2	3		5
8. Weight and Capacity	1		3		4
Total	8	12	12	8	40

Marks by thinking level

Each question is counted at its main thinking level.

Level	What the student does	Marks	Share
Remember	recall a fact or term	2	5%
Understand	explain or interpret an idea	2	5%
Apply	use an idea or rule in a new problem	25	62%
Analyse and create	trace, debug, compare, design or justify	11	28%

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Blueprint: Annual examination

Syllabus: all chapters, mostly chapters 9-15 (term 2). Marks by chapter and section.

Chapter	A	B	C	D	Marks
3. Angles as Turns	1				1
9. Coconut Farm	1	2	3		6
10. Symmetrical Designs	1	2			3
11. Grandmother's Quilt	1	2	3		6
12. Racing Seconds	1	2	3		6
13. Animal Jumps	1	2	3		6
14. Maps and Locations	1	2		4	7
15. Data Through Pictures	1			4	5
Total	8	12	12	8	40

Marks by thinking level

Each question is counted at its main thinking level.

Level	What the student does	Marks	Share
Remember	recall a fact or term	3	8%
Understand	explain or interpret an idea	3	8%
Apply	use an idea or rule in a new problem	19	48%
Analyse and create	trace, debug, compare, design or justify	15	38%

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Marking schemes

Accept any correct method and any sensible wording. Give part marks for correct steps, even when the final answer has a slip.

Marking scheme: Half-yearly examination

1. a) true
2. b) 12
3. b) 8
4. b) 90°
5. c) 120°
6. a) input
7. b) 15
8. b) halving search
9. 18 and 8 (1 mark for each)
10. West (2 marks; 1 mark for a correct method)
11. (a) > (b) = (1 mark for each)
12. 14 litres; 2½ jugs (1 mark for each)
13. 3 and 39 (1 mark for each)
14. 22; fuel = fuel + 12 (1 mark for each)
15. The numbers can be 14, 23, 32, 41 or 50. (a) true (b) can't tell (c) false (1 mark for each)
16. 50: 51 to 100 left. 75: 51 to 74 left. 62: 63 to 74 left. (Other middles, such as 63, are fine.) (1 mark for each guess with the numbers left)
17. tiles = step x step; 64; step 10 (1 mark for each)
18. 450, 450, 380, 380. The shop 380 m away. (2 marks for the values (1 mark with one slip), 1 mark for the answer)
19. (a) 9, 20, 28, 40, 50 (b) 5 (c) 5 cups ((a) 2 marks, 1 mark with one slip (b) 1 mark (c) 1 mark)
20. (a) 12 (b) 17 (c) 0 (d) If children = 40 then say 'Bus full', else open the door. (1 mark for each part)

Marking scheme: Annual examination

1. b) 22
2. b) 45°
3. a) tiling
4. c) 120
5. b) 30
6. b) 200 m
7. a) a picture graph
8. a) facing the same way as at the start
9. x 3, then + 1 (2 marks; 1 mark for a rule that fits one pair)
10. 4 and 3 (1 mark for each)
11. 12 square units (2 marks; 1 mark for 6 halves = 3)
12. 65, 58, 62; Om won (1 mark for the seconds, 1 mark for Om)
13. 10 and 20 (1 mark for each)
14. F4 (2 marks; 1 mark for C4 after going North)
15. (a) 17 (b) 14 (c) Yes, the outputs are different (1 mark for each)
16. It says 10, 7, 4, 1, -2, ... and never reaches exactly 0, so it never stops. Fix: repeat until timer < 1 (or take away 2, or start at 9). (1 mark for the trace, 1 mark for the bug, 1 mark for a fix)
17. 15 and 30 (1 mark for listing the landings, 1 mark for 15, 1 mark for 30)
18. 6 tiles; 6 red; 18 white (1 mark for each)
19. (a) D5 (b) For example East 3, North 4 (7 moves); any 7-move route that avoids A2 and B2 (c) 700 m ((a) 1 mark (b) 2 marks: 1 for directions, 1 for avoiding the walls (c) 1 mark)
20. (a) 10 (b) 12 (c) Bo (d) Open answer, for example: a bump should cost less than a coin, or the winner is the fastest player. (1 mark for each part)

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